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$$f(x, y) = 2xye^{\frac{-4x^2 + y^2}{8}}$$

$$\frac{\partial f}{\partial x} = 2ye^{\frac{-4x^2 + y^2}{8}} + 2xy \left(\frac{-8x}{8} \right) e^{\frac{-4x^2 + y^2}{8}} = 2ye^{\frac{-4x^2 + y^2}{8}} - 2x^2ye^{\frac{-4x^2 + y^2}{8}} =$$

$$2ye^{\frac{-4x^2 + y^2}{8}} (1 - x^2)$$

$$\frac{\partial f}{\partial y} = 2xe^{\frac{-4x^2 + y^2}{8}} + 2xy \left(\frac{-2y}{8} \right) e^{\frac{-4x^2 + y^2}{8}} = 2xe^{\frac{-4x^2 + y^2}{8}} - \frac{1}{2}xy^2e^{\frac{-4x^2 + y^2}{8}} =$$

$$2xe^{\frac{-4x^2 + y^2}{8}} \left(1 - \frac{y^2}{4} \right)$$

$$\Rightarrow y(1 - x^2) = 0 \rightarrow y = 0, x = \pm 1$$

$$\Rightarrow x \left(1 - \frac{y^2}{4} \right) = 0$$

$$\text{Als } y = 0 : x \left(1 - \frac{0^2}{4} \right) = 0$$

$$\Rightarrow x = 0$$

$$\text{Als } x = \pm 1 : 1 \left(1 - \frac{y^2}{4} \right) = 0$$

$$\Rightarrow 1 - \frac{y^2}{4} = 0$$

$$\Rightarrow -\frac{y^2}{4} = -1$$

$$\Rightarrow y^2 = 4$$

$$\Rightarrow y = \pm 2$$

$$\Rightarrow (0, 0), (1, 2), (-1, 2), (1, -2), (-1, -2)$$

References